Singapore rainfall: temporal-spatial patterns and prediction

Tasks 1 and 2: Spark exploration and probability prediction, 2017–2024

Group members. Pan Yuxiang, Hou Jiacheng, Wu Jiarui. **Scope and status.** This report covers Task 1 rainfall exploration and Task 2 half-hour exceedance probability prediction. Models have been trained, calibrated and evaluated on gpushare. Group members are Pan Yuxiang, Hou Jiacheng and Wu Jiarui.

Data and methods. The eight NEA / data.gov.sg annual datasets contain 48,538,713 source records and 48,538,627 unique valid observations from 91 station IDs. Spark performs full-data cleaning and aggregation. Annual comparisons use 21 stable stations to reduce changes caused by the monitoring network.

Main findings. Of all valid five-minute observations, 97.08% are dry. Eligible diurnal samples show the highest rain frequency at 16:00–17:00 Singapore time. Available-year monthly averages have the highest frequency in November and the lowest in July, with varying eligible years and stations. In the fixed panel, duration-normalized rainfall is lowest in 2019 and highest in 2021. Nearby stations generally have more strongly correlated daily rainfall; eight years alone do not establish long-term climate change.

Year	Rain / observed 24h (mm)	Wet frequency (%)	Wet-interval intensity (mm/5min)	Mean coverage (%)
2017	6.85	2.85	0.833	94.6
2018	7.10	2.90	0.840	82.9
2019	4.81	1.97	0.841	94.7
2020	7.40	2.92	0.877	95.3
2021	7.84	3.04	0.895	97.8
2022	7.71	3.08	0.868	99.0
2023	7.52	3.16	0.824	95.2
2024	7.11	3.01	0.819	95.6

Each year in this table uses the same 21 stations, with equal station weights. Duration-normalized rainfall is neither an official annual total nor an actual daily total. Eligible monthly and hourly samples can vary.

1. Data sources, fields and cleaning

Sources and fields. Records contain observation and update timestamps, station ID/name/device ID, coordinates, reading update timestamp, rainfall amount, type, unit and date fields. There are no temperature, humidity or wind observations. API previews identify fields and annual IDs; weather findings come from the complete annual files.

Source tracking. During execution, each full CSV receives a manifest containing its official dataset ID, API row count, retrieval time, byte count and SHA256 of the original CSV bytes. The read row count is reconciled against the recorded annual API count.

Cleaning. Spark parses timestamps in Asia/Singapore. It accepts finite, nonnegative mm readings with the expected five-minute type and valid station ID, timestamp and CSV record. Timestamps are aligned only if within two seconds of the five-minute grid; original epochs are retained. Station-time duplicates use the latest reading update, then deterministic tie-breakers. Combined annual data are checked again for cross-file duplicates.

Year	Source rows	Duplicates removed	Off-grid over 2 s	Negative readings	Stations
2017	5,256,106	86	0	0	62
2018	4,594,246	0	0	0	56
2019	5,026,862	0	0	0	54
2020	6,334,473	0	0	0	80
2021	7,044,646	0	0	0	76
2022	7,133,230	0	0	0	72
2023	6,742,890	0	0	0	74
2024	6,406,260	0	0	0	70

Cleaning results. All 86 duplicates occur in 2017. There are no negative readings or timestamps more than two seconds off-grid, and no cross-file duplicates. No station-year with at least 90% coverage has all-zero rainfall. Missing intervals are never zero-filled. Readings above 50 mm per five minutes are retained and flagged for sensitivity analysis; none occur in this run. This is a screening value rather than an official quality threshold; valid records can still contain sensor errors.

2. Metrics and comparable samples

For a station and time period, let N be the number of valid observations, $r_i \geq 0$ the rainfall amount, $W = \sum_i \mathbf{1}(r_i > 0)$ the number of wet intervals, and E the expected calendar slots. We use:

$$\begin{aligned} \text{Coverage} &= N/E, \qquad E = 288 \times \text{calendar days}, \\ \text{Wet frequency (\%)} &= 100W/N, \\ \text{Wet-interval intensity (mm/5min)} &= \sum_i r_i/W \quad (W > 0), \\ \text{Rain / observed 24h (mm)} &= 288 \sum_i r_i/N. \end{aligned}$$

Calendar denominators handle leap years. Low coverage can reflect a station operating for only part of a year. Intensity averages wet intervals only. For one station and period, normalized rainfall equals 288 × frequency/100 × wet-interval intensity. This product generally does not hold after separate station averaging. Normalized rainfall is not an official daily or annual total and does not reconstruct missed storms.

Fixed annual panel. Of 91 station IDs across eight years, 21 satisfy all criteria: observations in every year, annual coverage of at least 80% in each year, and latitude/longitude spans no larger than 0.005°. The other 70 are excluded only from fixed-panel annual comparisons; their valid observations were not deleted during cleaning. The 21 stations receive equal weights. The next page lists the station scope for each analysis.

Monthly and diurnal samples. Figures 2 and 4 further screen sufficiently complete months or hourly strata from the 21-station pool. Monthly summaries average stations equally, then available years equally. Each eligible year-month has 17–21 stations, and each month has 6–8 available years. Requiring the same station-month to qualify in all eight years leaves only May, June, July, September, October and December, used as an additional comparison.

Spatial samples. Figure 1 uses 2024, which has 70 stations. Of these, 56 have annual coverage of at least 90% and enter map colouring; the other 14 are shown in grey. Figure 5 uses daily totals from the fixed 21 stations, requiring at least 95% daily coverage and 365 common valid days per pair. The 80%, 90% and 95% completeness thresholds are project choices.

3. Computing environment and station scope

Actual computing environment. The main Task 1 analysis ran on an Apple Silicon Mac: 10 logical CPUs, 16 GiB RAM, macOS 26.5.2 ARM64, Python 3.12.14, Java Temurin 17.0.20.1+1 and Spark 3.5.8. Task 2 labels, features, all models and evaluations ran on gpushare on 9 October 2026: Python 3.11.10, Azul Zulu Java 17.0.20.1, Spark 3.5.8 and XGBoost 3.2.0. The container quota is 42 CPUs and about 667.68 GiB RAM; the project binds only CPUs 0 and 1, uses a 2 GiB driver heap and a sampled 4 GiB process-tree RSS stop bound.

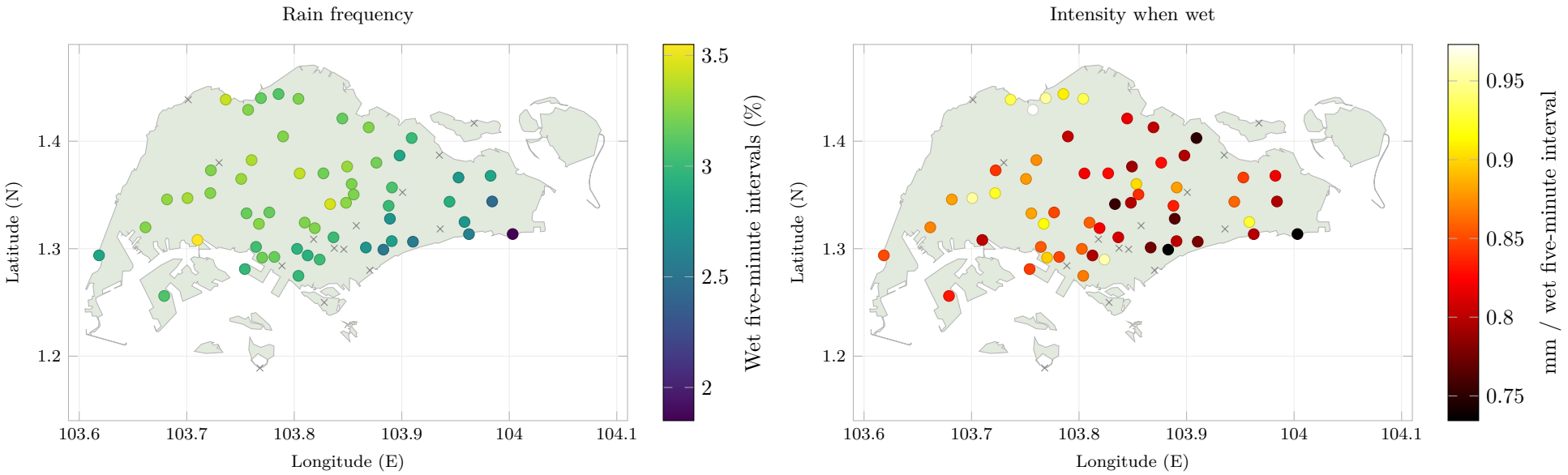
Spark configuration and role. Spark uses `local[2]`: two worker threads on one machine. Driver heap is 2 GiB, shuffle partitions 8 and default parallelism 2. Annual processing is sequential and uses no GPU. Spark DataFrame/SQL and windows perform cleaning, deduplication, temporal grouping and full-data aggregation; small summaries are used for figures and daily-rainfall correlations.

Which stations enter each analysis? Screening applies to a specific comparison; it does not remove all stations outside the fixed panel.

Analysis	Station scope	Screening rule
Figure 1: 2024 station map	56 of 70 stations that year	Annual coverage $\geq 90\%$; 14 shown in grey
Figure 2: monthly patterns	Eligible subsets of the fixed 21	17–21 stations per eligible year-month
Figure 3: annual changes	The same 21 stations	Other 70 excluded from this comparison only
Figure 4: diurnal patterns	Eligible strata of the fixed 21	Year-month-hour coverage $\geq 80\%$
Figure 5: distance and correlation	Fixed 21 stations; 210 pairs	Daily coverage $\geq 95\%$; ≥ 365 shared days
Figure 6 left: wet amounts	All 91 stations across eight years	Wet five-minute observations only
Figure 6 middle/right: exceedance and persistence	Valid 2024 observations from all 70 stations	Six complete future slots or two adjacent slots

Cleaning checks. Source row counts match API counts; 86 duplicate records were removed, and missing slots are not zero-filled. The overall dry fraction uses all 91 stations, rather than only the 21-station panel.

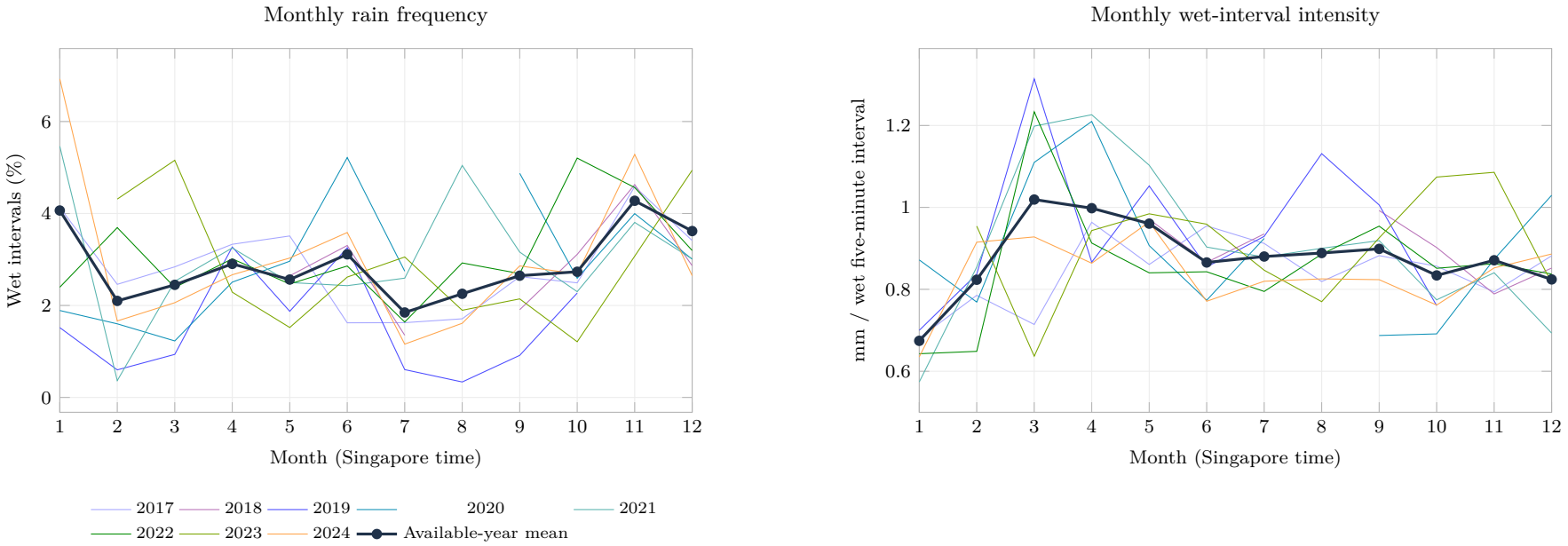
Figure 1: Station rainfall maps in 2024



In 2024, 56 of 70 stations have annual coverage of at least 90%. Coloured points show individual gauge statistics; grey crosses indicate lower coverage. S33 (Jurong Pier Road) has the highest rain frequency, 3.55%. No spatial interpolation is performed. The geoBoundaries land outline is derived from URA 2016 data under ODbL 1.0 and serves as a geographic reference.

Task 2 feature implications (stations). Candidate features include station ID and coordinates to represent station differences. Task 2 should screen windows for historical-feature and future-label completeness, rather than automatically restrict training to the 21-station annual panel.

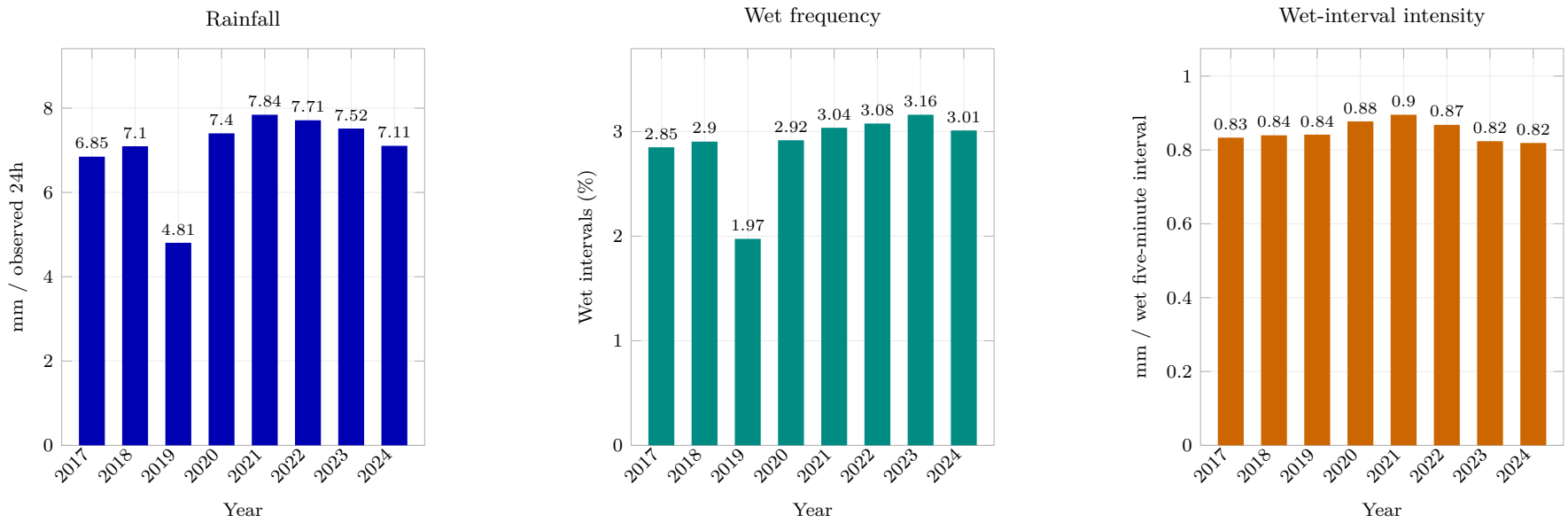
Figure 2: Monthly patterns within a year



Eligible station-year-months from the fixed annual pool require at least 80% coverage. Stations are averaged equally, then available years equally. Mean frequency is highest in November (4.28%) and lowest in July (1.85%). Mean wet-interval intensity is about 1.02 mm in March and 0.67 mm in January. Frequency and intensity need not share the same seasonal pattern. Each month has 6–8 available years and varying eligible stations. MSS descriptions of wetter November–January conditions provide context rather than proof of monsoon causation.

Task 2 feature implications (months). Retain cyclic month or day-of-year features and test their contribution by ablation. Singapore is affected by monsoons but has no sharply distinct wet and dry seasons. Less frequent rain does not mean no rain, and frequency is different from accumulated rainfall.

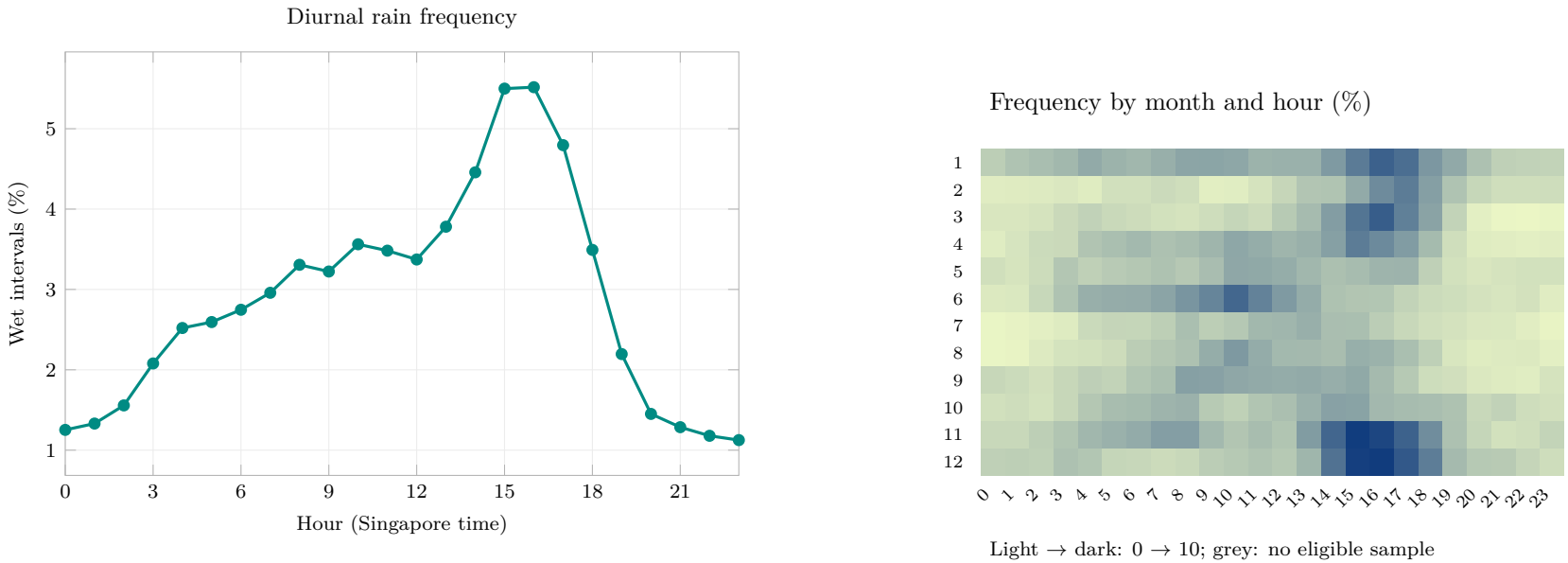
Figure 3: Annual changes in the fixed panel



The same 21 stations enter every year, each with at least 80% annual coverage. Mean rainfall per observed 24 hours ranges from 4.81 mm in 2019 to 7.84 mm in 2021. Frequency and wet-interval intensity change differently, helping explain annual differences. Coverage normalization does not recover missed storms; mean coverage in 2018 is about 82.9%, lower than in most years. These are descriptive annual comparisons, without a long-term climate trend or significance claim.

Task 2 feature implications (years). Initially omit calendar-year identifiers from the main model to reduce memorization of individual years. NEA documented dry conditions and a strong positive Indian Ocean Dipole in 2019; this is external climate context rather than causal evidence from these data. Use chronological training, validation and test splits, and report performance across years.

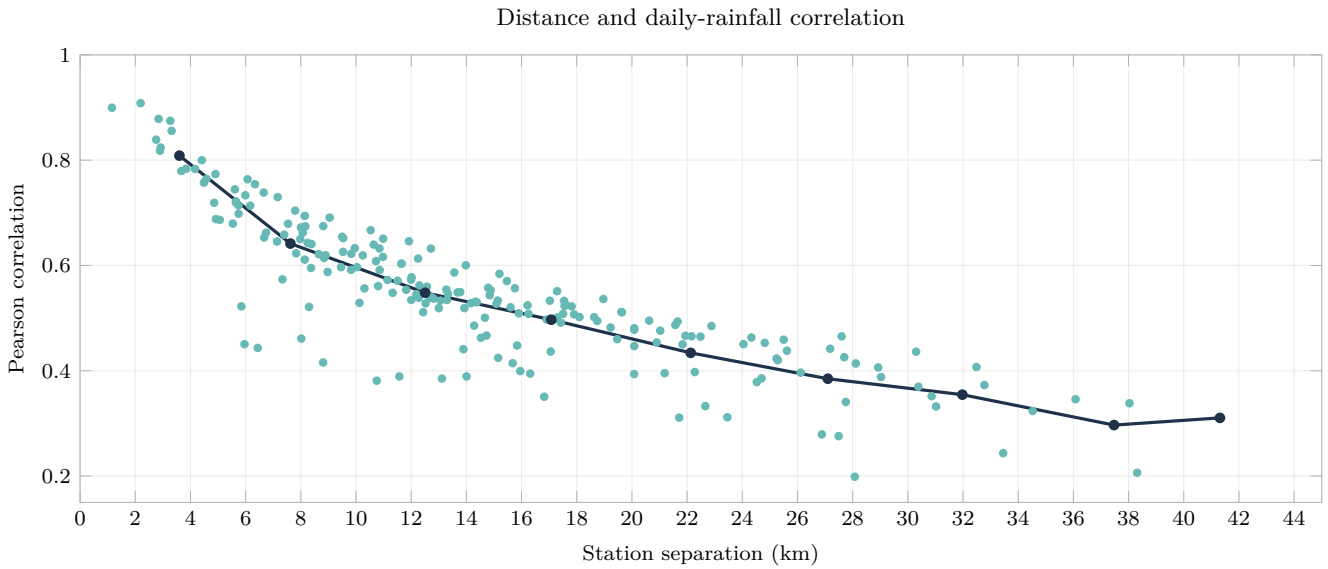
Figure 4: When during the day does it rain?



Frequency peaks at 16:00–17:00 Singapore time. The right panel shows how the cycle varies by month; June has elevated frequencies closer to morning and midday. Eligible station-year-month-hour strata require at least 80% coverage and receive equal weights; their composition can vary. MSS afternoon-rainfall descriptions provide background. No additional temperature, wind or humidity observations are available to explain mechanisms.

Task 2 feature implications (hour and month). Encode Singapore local hour cyclically and consider month–hour interactions: the strength and timing of the afternoon peak vary by month. Logistic regression and linear SVM need explicit interaction features; trees can learn such combinations. Their predictive contribution must still be tested by ablation.

Figure 5: Distance and daily-rainfall correlation

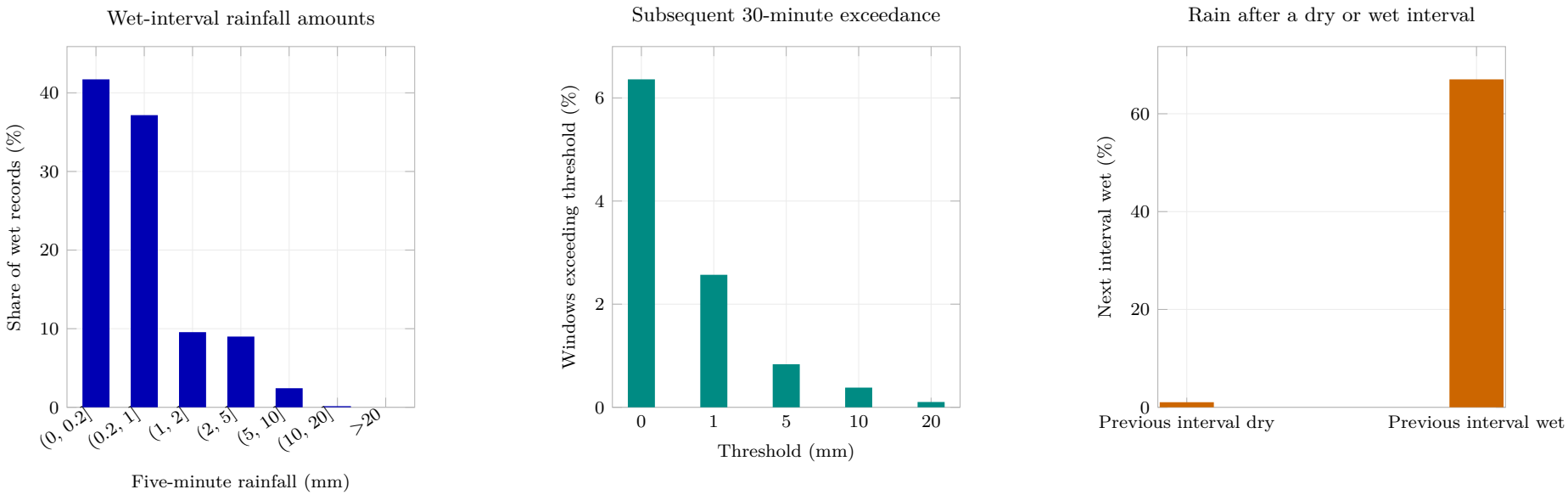


Each point is a station pair: separation is on the horizontal axis and Pearson correlation of daily totals on shared valid dates is on the vertical axis. The dark line averages correlations within 5 km distance groups. The 21 stations form 210 pairs. Mean correlation is 0.809 for 17 pairs no more than 5 km apart, and 0.336 for 12 pairs more than 30 km apart. **Nearby stations generally show stronger daily-rainfall co-movement.** This is an overall pattern, not an ordering that holds for every pair.

Stations are not independent. Nearby gauges may experience the same rainfall event, and different pairs can share a station. Thus, 91 gauges are not 91 independent weather samples, and the 210 correlations are not 210 independent samples. Shared-day screening follows the criteria given earlier.

Task 2 feature implications (neighbours). Use available history from the three closest stations: means, maxima, wet fractions, available-neighbour counts, distances and reading ages. Graph neural networks are deferred. Daily correlation motivates these candidates; Task 2 separately compares models with and without neighbour features to assess predictive gains.

Figure 6: Rainfall amounts, half-hour exceedance and persistence



Left: rainfall amounts when wet. The denominator contains **wet** five-minute records from all stations across eight years. Bars show the proportions in rainfall ranges: 41.72% are in $(0, 0.2]$ mm and 37.16% in $(0.2, 1]$ mm. Together, about 78.88% of wet intervals have no more than 1 mm. Dry records are excluded from this panel; they represent 97.08% of all valid observations.

Middle: how often a half-hour total exceeds a threshold. For 6,374,476 complete 2024 windows, the sum of six five-minute readings in $(t, t + 30\text{min}]$ defines the subsequent total. Sums are rounded to nine decimal places to avoid floating-point errors at exact thresholds. The horizontal axis lists thresholds and the vertical axis shows the fraction strictly exceeding each threshold. Totals exceed 0 mm in 6.36% of windows and 5 mm in 0.83%. Larger thresholds are rarer. The five categories have equal display spacing, not equal numerical spacing.

Right: does rainfall continue? For adjacent five-minute intervals at the same station in 2024, a wet interval follows a dry one in 1.04% of cases, versus 67.03% after a wet one. Recent rainfall therefore motivates features for Task 2. The middle and right panels are historical statistics, **not trained-model performance**.

Task 2 feature implications (persistence and rare events). Implemented features include the latest reading, 15/30/60-minute totals, recent rainfall changes and wet-spell duration. Only records available at prediction time are used; missing readings are not dry readings. The following sections report positive counts, PR-AUC, Brier score and calibration curves separately by threshold.

4. Task 2: prediction target and chronological splits

Prediction target. Given station s, Singapore prediction time t and rainfall threshold q, return the probability that the next 30-minute accumulated rainfall strictly exceeds q. The interface retrieves station and neighbour history as of t; users do not manually supply an observation sequence.

$$R^{30}_{s,t} = \sum_{k=1}^{6} r_{s,t+5k\,\min}, \qquad y^{(q)}_{s,t} = \mathbf{1}(R^{30}_{s,t} > q), \qquad \hat{p}_q = P(y^{(q)} = 1 \mid x_{s,t}).$$

Thresholds and interface scope. This experiment uses $q \in \{0, 5, 25\}$ mm: 0 denotes any rain, 5 a larger half-hour accumulation, and 25 a rare high accumulation. These are project thresholds, not official Singapore storm categories. Three classifiers are trained separately; each station-time row stores one feature set and three target columns, without triplicating the feature dataset. The input threshold selects a trained model. Arbitrary untrained thresholds are currently unsupported; a requirement for continuous thresholds would need a distributional or threshold-conditioned extension. The 50 mm and graph neural network experiments are deferred.

Valid labels. Require exactly six distinct valid future five-minute slots; the current reading is excluded. Round totals to nine decimal places before applying strict exceedance, so equality is negative. A missing future slot invalidates the window; missing rain is not filled with zero. Treating reading timestamps as interval ends is an operational assumption; the archive does not establish real-time first-publication times.

Purpose	Valid rows	> 0 positives	> 5 positives	> 25 positives
2017–2022: eligible training pool	34,542,285	–	–	–
2017–2022: fixed 5% training sample	1,727,966	103,477	13,712	926
2023 H1: probability calibration	3,325,499	251,794	30,699	1,532
2023 H2: model selection	3,393,373	197,358	28,000	2,487
2024: final test	6,374,476	405,430	53,193	3,517

Temporal leakage control. Label windows cannot cross training, calibration, selection or test boundaries. The July split purges 402 additional rows. A deterministic hash sample with seed 5208 selects 5% of eligible real training rows, shared by all models. Training is resource-limited, not full-data training. Validation and test retain their complete natural distributions. The full feature table contains 47,636,035 rows. Since Task 1 explored 2024, it is a later chronological test excluded from model selection, rather than data never inspected anywhere in the study.

5. From Task 1 findings to feature engineering

Task 1 motivation	Construction	Implemented inputs
Afternoon peak varies by month	Cyclic local hour/month and four interactions	8 calendar features; no year ID
Station differences and spatial correlation	Station ID, coordinates, three closest neighbours	Neighbour means, maxima, wet fractions
Short-term rainfall persistence	Latest reading and 5/10/15/30/60-min lags	15/30/60-min totals, changes, wet-spell length
Missingness and unequal coverage	Valid counts, availability and reading age	Missing flags, available-neighbour counts/distances

As-of availability. Search the preceding 90 minutes and require observation, aligned-slot and stored-update times to be no later than t. A latest reading older than 30 minutes is unavailable. Rolling totals end at the latest eligible reading and include its age and valid-window counts. Exact lags use calendar slots; the preceding row after a gap is not a five-minute lag. Stored archive update times are conservative availability proxies, not first-publication timestamps.

Neighbour construction. Choose the three closest other stations using coordinates known by prediction time. Disable a station spatial contribution if it moves more than 0.005° from its initial location. Join only readings available as of t. Aggregate latest amounts and past 30-minute totals into means, maxima and available counts, plus wet fractions, mean distances and ages. Spark handles full-data time windows, features and neighbour joins; a small coordinate table supports distance calculations.

Preprocessing and dimensions. Explicitly list 46 numeric candidates and station ID, excluding future totals, labels and audit columns. Omit the constant geometry-change value in the fixed training sample, retaining 45 values and 27 informative missing flags. Mean imputation is fitted only on the training sample. Linear models use station one-hot encoding, giving 162 dimensions. Trees use station indices with categorical metadata, giving 73 dimensions. XGBoost explicitly marks station as categorical rather than treating its index as a continuous ordering.

L2 logistic feature ablation	Test PR-AUC: 0 mm	5 mm	25 mm
Calendar and station	0.11652	0.01647	0.00136
Add own-station history	0.53061	0.15788	0.01485
Add neighbours	0.55978	0.17909	0.01734

Ablation interpretation. The initial nine natural-distribution logistic models share the training sample, use full 2023 validation, have no subsequent Platt calibration and use an 80-iteration cap. Own-station history supplies the largest gain; neighbours further improve PR-AUC for all targets. The later comparison therefore retains the full feature set. This supports feature groups, not the usefulness of every individual variable. Later parameters are not reselected from test results.

6. Models, imbalance policies and probability calibration

Model	Fixed parameters
L2 logistic (MLlib)	$\lambda = 0.001$, standardization, 100 iterations, tolerance 10^{-6}
Linear SVM (MLlib)	$\lambda = 0.001$, standardization, 100 iterations, tolerance 10^{-6}
GBT (MLlib)	40 rounds, depth 4, step 0.1, 128 bins, minimum 20 rows/node
XGBoost (Spark interface)	80 trees, depth 4, learning rate 0.1, 64 bins, child weight 20; $L_2 = 1$

Spark role. The first three algorithms use `pyspark.ml`, the MLlib DataFrame API. XGBoost uses the external `SparkXGBClassifier`, not an MLlib built-in. Training, prediction and full-data evaluation run through Spark; XGBoost uses two CPU workers. Parameters are fixed in advance with seed 5208, without a systematic hyperparameter-optimization claim. All nine SVM and two weighted logistic candidates reach the 100-iteration cap; additional iterations could change the comparison.

Three training policies. Natural distribution assigns unit weights. Balanced weighting uses $w_+ = N/(2P)$ and $w_- = N/[2(N-P)]$, giving equal class mass and overall mean weight one. Constrained SMOTE targets 20% positives, caps their total at ten times the original positives, and augments training only. Calibration, selection and test are neither resampled nor class-weighted.

Why constrain SMOTE? Unrestricted interpolation can create invalid station encodings, broken cyclic calendars or contradictory wet counts. Here, search up to five minority neighbours only within the same station, missingness, discrete-state and rain-zero pattern. Interpolate only 16 continuous rain features with one shared coefficient. Calendar, coordinates, availability counts, wet fractions and ages remain fixed; isolated profiles fall back to duplication. Augmented vectors pass nonnegativity, cyclic-norm, complete-total ordering and wet-count consistency checks. Single-thread sklearn performs minority-neighbour search in a bounded training pool; Spark performs the remaining data and model operations.

Threshold mm	Added rows	Augmented positives (%)	Rows with changed rain features	Isolated-profile copies
0	302,646	20.000	129,231	73,685
5	123,408	7.406	34,344	48,033
25	8,334	0.533	773	6,061

Calibration. Fit a Platt map for each candidate on naturally distributed 2023 H1: $\hat{p} = \sigma(az + b)$ with $a \geq 0$, using MLlib logistic regression with regularization 10^{-6}. This converts SVM margins into probabilities and adjusts probability scales affected by weighting or SMOTE. Calibration does not guarantee accurate probabilities in a later year.

7. Evaluation protocol and validation selection

Primary metrics. ROC-AUC measures positive-negative ranking; random ranking is about 0.5. PR-AUC emphasizes positive precision and recall and is more informative for rare events. We use Spark trapezoidal PR integration over exact score ranks (`numBins=0`), not sklearn average precision. Uninformative ranking has a reference PR level near prevalence: 0.06360, 0.008345 and 0.0005517 for the three test targets.

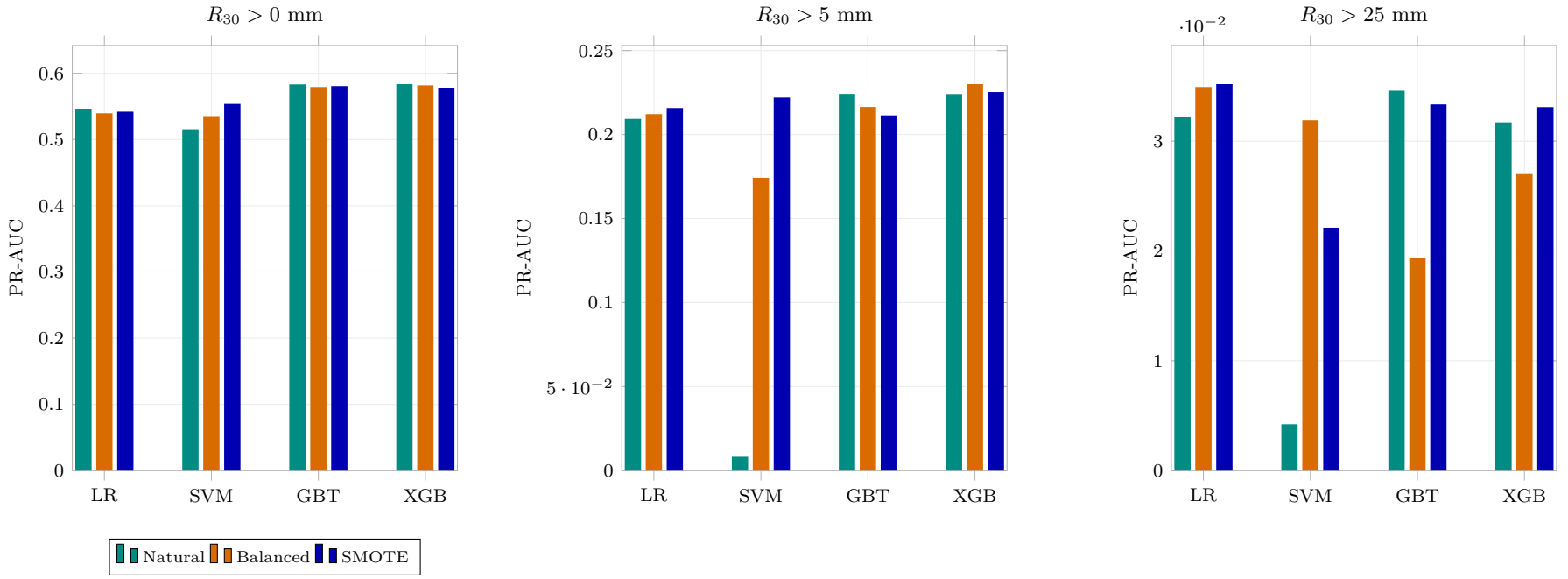
$$\text{Brier} = \frac{1}{n}\sum_{i=1}^{n}(\hat{p}_i - y_i)^2, \qquad \text{Precision} = \frac{TP}{TP+FP}, \quad \text{Recall} = \frac{TP}{TP+FN}.$$

Probability quality and point classification. Lower Brier is better, compared against a constant prediction equal to real-training prevalence. Reliability curves compare mean predictions with observed frequencies in approximately equal-mass bins; score ties merge groups, so there need not be exactly 20 points. Precision, recall, F1 and accuracy are supplementary. Choose a probability cutoff from a candidate grid by selection-set F1, then freeze it. This classification cutoff differs from the input rainfall threshold q.

Algorithm / policy	Selection PR-AUC: 0 mm	5 mm	25 mm
Logistic / Natural	0.54547	0.20932	0.03221
Logistic / Balanced	0.53968	0.21216	0.03493
Logistic / SMOTE	0.54213	0.21581	0.03520
SVM / Natural	0.51546	0.00825	0.00422
SVM / Balanced	0.53541	0.17428	0.03191
SVM / SMOTE	0.55375	0.22202	0.02212
GBT / Natural	0.58332	0.22421	0.03460
GBT / Balanced	0.57932	0.21642	0.01934
GBT / SMOTE	0.58073	0.21141	0.03335
XGBoost / Natural	0.58368	0.22413	0.03171
XGBoost / Balanced	0.58186	0.23002	0.02701
XGBoost / SMOTE	0.57796	0.22531	0.03310

Frozen selection. Choose the policy for each algorithm and threshold by 2023 H2 PR-AUC, using Brier only to break ties, then choose the default algorithm. Default routing is natural XGBoost for 0 mm, balanced XGBoost for 5 mm, and SMOTE logistic for 25 mm. Evaluate the 12 selected algorithm-threshold models on full 2024. SMOTE improves some linear candidates, but GBT selects natural training for all targets; it does not universally improve models.

Figure 7: imbalance policies on the selection set



Reading the figure. Each panel represents one rainfall threshold. Three bars per algorithm compare training policies on the complete naturally distributed selection set. Higher thresholds are rarer; panels use separate vertical scales. Trees lead at 0 mm, balanced XGBoost is best at 5 mm, and SMOTE logistic has the highest selection PR-AUC at 25 mm. Policy effectiveness depends on algorithm and target, not simply on increasing training positives.

8. Chronological test results in 2024

Rainfall threshold mm	Algorithm	Selected policy	ROC-AUC	PR-AUC	Brier
0	Logistic	Natural	0.85500	0.55978	0.0392757
0	SVM	SMOTE	0.86049	0.57387	0.0399706
0	GBT	Natural	0.87749	0.60338	0.0368843
0	XGBoost	Natural	0.87986	0.60797	0.0364959
5	Logistic	SMOTE	0.85004	0.18162	0.0075438
5	SVM	SMOTE	0.77601	0.20093	0.0077220
5	GBT	Natural	0.86100	0.20891	0.0074329
5	XGBoost	Balanced	0.87526	0.21299	0.0072292
25	Logistic	SMOTE	0.85651	0.01805	0.0005483
25	SVM	Balanced	0.84840	0.01550	0.0005484
25	GBT	Natural	0.76516	0.01881	0.0005618
25	XGBoost	SMOTE	0.88889	0.01918	0.0005461

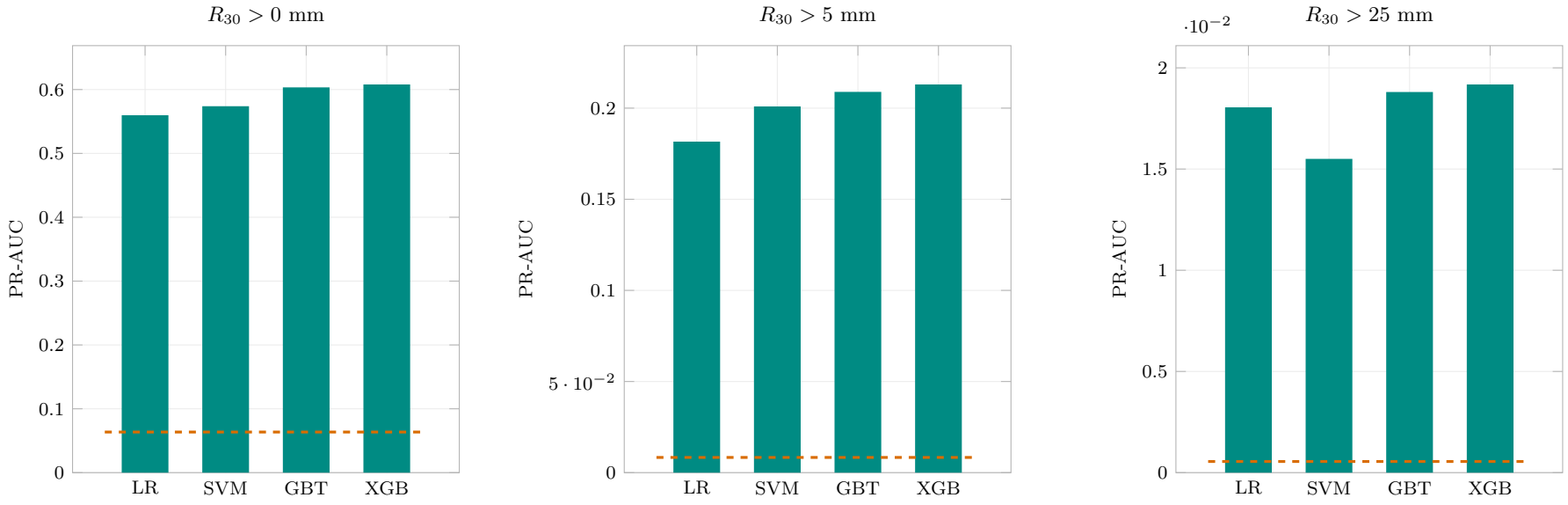
Test size and baselines. Each target is evaluated on 6,374,476 real windows with prevalences of 6.3602%, 0.83447% and 0.055173%. Constant training-prevalence Brier baselines are 0.0595707, 0.0082752 and 0.0005514. Always predicting negative gives accuracies of 93.64%, 99.17% and 99.945%, showing why accuracy cannot replace rare-event detection. All table results use calibrated probabilities.

Interpretation. XGBoost test PR-AUC is 0.60797, 0.21299 and 0.01918 for the three targets. At 25 mm, ROC-AUC remains 0.88889 but PR-AUC is low: rare events can be enriched, yet reliable detection remains difficult. The 25 mm GBT Brier of 0.0005618 is worse than the constant baseline. The default 25 mm model remains validation-selected logistic; it is not switched to XGBoost after inspecting test results.

Default target mm	Algorithm	Training diagnostic PR-AUC	Selection PR-AUC	Test PR-AUC
0	XGB	0.61237	0.58368	0.60797
5	XGB	0.22789	0.23002	0.21299
25	LOGISTIC	0.04515	0.03520	0.01805

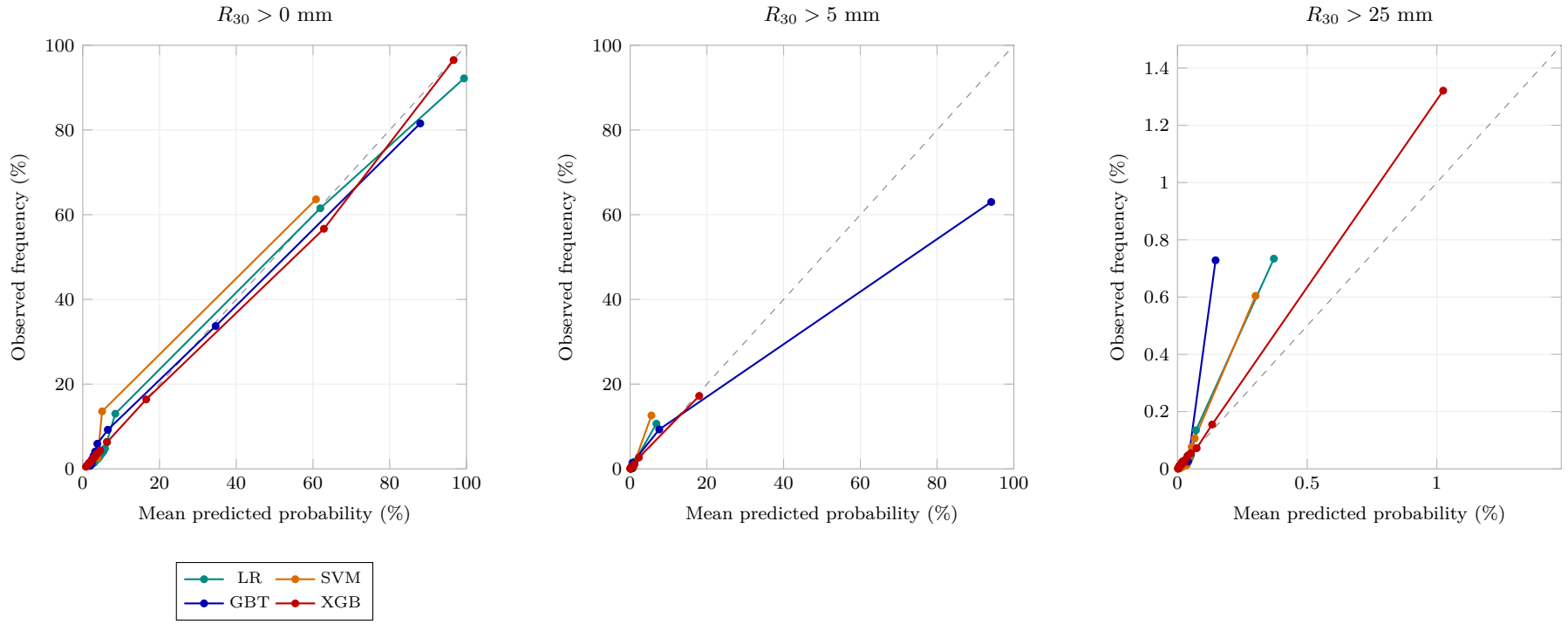
In-sample and out-of-sample evaluation. Training diagnostics use the same deterministic 20% subset of original real training rows (344,957 rows), excluding synthetic rows. Selection and test are evaluated in full. Their prevalences differ, so PR-AUC gaps are not entirely attributable to overfitting. Natural GBT at 25 mm has training diagnostic PR-AUC 0.17452 versus selection 0.03460, indicating rare-label generalization difficulty. The 926 original training positives can be overlapping windows, not 926 independent storms.

Figure 8: test ranking and probability error



Reference lines. Orange dashed lines show test prevalence, an uninformative-ranking PR reference. Bars use the frozen algorithm-specific policies from the preceding table. Although 25 mm results exceed the roughly 0.00055 reference, absolute PR-AUC remains small. High ROC-AUC does not imply high precision for extreme rain. Probability errors are assessed in the Brier table and the next reliability plots.

Figure 9: reliability curves and probability quality



Reading the curves. The grey diagonal denotes ideal calibration. Points above it indicate underprediction; below it, overprediction. Each point groups real test windows with similar scores; fewer points do not mean fewer test examples. For rare events most probabilities are near zero, and even the highest-probability group includes many low-risk windows. These curves do not fully diagnose the high-risk tail.

Observed calibration benefit. For SMOTE logistic at 25 mm, selection Brier falls from 0.0011352 before calibration to 0.0007248 afterward. Mean probability falls from 0.0055317 to 0.0004886, against actual prevalence 0.0007329. Calibration substantially corrects resampling-induced inflation but still underpredicts some risk; later test evaluation remains necessary.

9. Default-model point metrics and observations

Rainfall threshold mm	Validation-selected probability cutoff	Precision	Recall	F1	Accuracy
0	0.31989	0.61636	0.56524	0.58970	0.94997
5	0.15488	0.28201	0.34305	0.30955	0.98723
25	0.01462	0.05165	0.11288	0.07087	0.99837

Operational interpretation. These probability cutoffs illustrate one frozen operating point. The 25 mm F1 remains low, while high accuracy is largely due to negatives. Operational warnings could choose cutoffs based on false-alarm and missed-event costs, but optimizing cutoffs against test labels would not measure generalization. Main comparisons retain ROC-AUC and PR-AUC, with separate probability-quality assessment.

Information supplied by SMOTE. At 25 mm, 8,334 rows are added, but only 773 change rain features and 6,061 are isolated-profile copies. Constrained synthesis preserves consistency but cannot create new storm information. This helps explain why augmentation does not produce universal large improvements. Further work could enlarge real-positive training, tune rare-event weights and parameters, and evaluate by independent weather events.

Cross-threshold consistency. Independent training and calibration do not ensure $\hat{p}_{25} \leq \hat{p}_5 \leq \hat{p}_0$. Across full 2024 test, selected policies within each algorithm violate this ordering for logistic: 1, SVM: 2,448, GBT: 528 and XGBoost: 247 rows, each out of 6,374,476. These checks concern same-algorithm combinations; the mixed-algorithm default route has not undergone this full-data check. Joint monotone calibration is a possible extension. The present models do not constitute a fully consistent rainfall distribution.

Spatial and temporal dependence. Nearby gauges, shared events and overlapping half-hour windows produce dependent samples. Millions of rows are not millions of independent weather events. We report descriptive test performance without confidence intervals based on independent rows. Date- or event-block resampling could assess uncertainty and further examine transfer across stations and years.

10. Computing resources, saved models and prediction

Actual server configuration. gpushare runs single-machine Spark `local[2]` with CPU affinity 0 and 1, a 2 GiB driver heap, 8 shuffle partitions, single-thread BLAS/OpenMP, nice 19 and idle I/O priority, without GPU. Process-tree RSS is sampled every five seconds; exceeding 4 GiB or leaving less than 5 GiB project disk stops the job. This is sampled protection, not a hard OS quota. Benchmark sampled peak RSS is 2.719 GiB and minimum free disk 26.984 GiB, including interrupted attempts. This deployment is not a multi-node cluster and does not simulate one with Docker.

Timing and scope. The 36 final valid base-model fits sum to about 32.05 minutes. Group worker times, including calibration, training diagnostics and selection evaluation, sum to about 58.30 minutes. The latest resumed invocation took about 48.95 minutes and reused completed groups, so it is not an end-to-end fresh-run time. Full-data label/feature preparation is separate; initial downloads, failed retries, artifact transfer and report compilation are outside the fit totals.

Saved models. Retain all 36 comparison pipelines and the initial nine logistic models, including fitted imputation, encoding, base classifier and Platt mapping. `models/model_comparison/manifest.json` records threshold routing, validation selection, parameters and synthesis version. The 36 comparison models contain 2,053 files and 3,869,936 bytes. Their internal metadata/data are required model artifacts, not removable statistical intermediates. Server and local model files were individually SHA256-verified.

Save/load checks. The 12 selected full pipelines pass save/load prediction checks on 15 label-free real feature rows, including three unknown station IDs. Probability differences from separate preprocessing/model steps are zero. An archive query using only station, time and threshold, S07 at 2024-03-01 12:00:00+08:00 with 5 mm, returns 0.00618169 (about 0.618%) from default balanced XGBoost. This verifies archive retrieval and model routing, not a live online weather service.

```
PYTHONPATH=.runtime/python python scripts/run_task2_models.py \
  --predict-station S07 --predict-time 2024-03-01T12:00:00+08:00 \
  --threshold 5
```

Material organization. README gives environment preparation, eight-year downloads, Spark features, training, evaluation and prediction commands. Raw annual files and working directories are excluded from Git and can be downloaded again from official sources. Reports and models can be viewed and loaded directly; generating features for new historical predictions requires rebuilding the data. Standalone LaTeX and models are retained for Git backup and synchronization to the Desktop copy.

11. AI use, sources and reproducibility

AI-use declaration. OpenAI Codex assisted with planning, API inspection, runtime setup, PySpark cleaning, aggregation, labels and feature engineering, MLlib and XGBoost training, constrained SMOTE implementation, execution checks, metric and plotting code, bilingual LaTeX writing and translation, compilation and layout review. Weather statistics and model results come from actual runs on real annual data. SMOTE is used only in training; synthetic vectors are not presented as observations or used as validation/test examples. Group members must review the code, methods and results and take responsibility for submission.

Sources.

1. NEA, Historical Rainfall across Singapore, annual datasets 2017–2024: data.gov.sg collection 2279. Source and checksum records are generated during downloads.
2. Meteorological Service Singapore, Climate of Singapore. Used only for seasonal and diurnal context, not as a replacement for project observations or proof of causation.
3. NEA, Review of the climate in 2019. External context on dry conditions and the positive Indian Ocean Dipole, not an input feature in the proposed model.
4. geoBoundaries, Singapore ADM0 metadata. The URA-derived outline represents 2016, under ODbL 1.0. Release metadata and GeoJSON are retained in `reports/assets/`; they are not a current coastline survey.
5. Apache Spark 3.5.8, MLlib classification and regression documentation, for logistic, linear SVM and GBT implementation. Evaluation uses Spark BinaryClassificationEvaluator.
6. XGBoost, PySpark estimator documentation, for external Spark integration; the actual installed version is 3.2.0.

Reproducibility. Source code and instructions are in `scripts/` and `README.md`. Both reports retain standalone LaTeX and PDF files. `models/` contains trained Spark pipelines, preprocessing and a threshold-routing manifest. Chart values are embedded directly in LaTeX, without separate images, statistical CSV files or interactive pages. Scripts regenerate statistics and checks in temporary working directories.

Submission status. Materials comprise complete source code with README, trained models and a PDF report, due 18 October 2026. Both analysis and modelling tasks are complete. This report lists member names only, as requested by the group. Git backup: `https://github.com/Panyuxiang0830/singapore_rainfall.git`. The server project is cleaned only after full-project completion and verification of durable backups.